

CREDITWISE: Intelligent Loan Approval System Using Machine Learning

Abstract

This paper presents an Intelligent Loan Approval System using Machine Learning. The system automates the loan eligibility decision by analyzing applicant details such as income, loan amount, and credit history. The proposed model reduces human bias and improves decision-making efficiency.

Introduction

Loan approval in financial institutions is traditionally a manual and time-consuming process. Machine learning provides an automated, data-driven solution to predict loan approval outcomes using historical data.

Methodology

The system uses a structured dataset containing applicant details. Data preprocessing is performed to handle missing values and encode categorical variables. The processed dataset is split into training and testing sets. A classification algorithm is trained using historical loan data to predict loan approval status.

Data Preprocessing

The dataset is cleaned by removing or replacing missing values using statistical methods such as mean and mode. Categorical attributes are converted into numerical values to make them suitable for machine learning algorithms.

Feature Selection

Independent variables such as income and loan amount are selected as input features, while loan status is considered the target output variable.

Model Training and Prediction

The classification model is trained using training data. The trained model predicts loan approval results for unseen data.

Performance Evaluation

The system performance is evaluated using accuracy and confusion matrix. These metrics measure the reliability and effectiveness of the prediction model.

Result

The system successfully predicts whether a loan is approved or rejected based on applicant details. The results demonstrate improved efficiency and automation in loan approval decision-making.

Conclusion

The proposed Intelligent Loan Approval System reduces manual effort and enhances prediction accuracy. This system can be integrated into banking applications for real-time loan eligibility checking.